

## HOMEWORK 14

Due date: Monday of Week 15,  
Exercises: 2.1, 2.2, 2.3, 2.7, 2.13, 2.14, 2.17, 2.18, 3.2, 3.3, 5.5, 5.7, 5.11, 5.12, 6.1, 6.2, 6.4, 6.5 pages 221-223.

**Problem 1.** Let  $G$  be a finite group with  $|G| = n$ . Let  $\iota : G \rightarrow S_n$  be the embedding in Cayley's theorem (namely,  $\iota : G \rightarrow \text{Perm}(G)$  is determined by the left multiplication action  $G \times G \rightarrow G$ ,  $(g, x) \mapsto gx$ ).

- (1) Let  $h \in G$  be an element of order  $d$  (so that  $d|n$ ). Show that  $\iota(h)$  is a product of  $n/d$  disjoint cycles of length  $d$ .
- (2) Suppose  $n = 4m + 2$ . Show that  $\iota(h) \in A_n$  if and only if the order of  $h$  is odd.

Hint: Let  $H = \langle h \rangle \subset G$  and consider the right coset decomposition  $G = \coprod Hg_i$ . For (2), see Ex. 10.1, page 74.

**Problem 2.** Let  $C$  be a conjugacy class of  $S_n$ . Decompose  $C \cap A_n$  as conjugacy classes of  $A_n$ .

This is roughly Ex.5.11. For future references, here is a more precise statement. Suppose that  $\sigma \in S_n$  has cycle lengths  $k_1, \dots, k_m$  with  $k_1 + \dots + k_m = n$ . This means that  $\sigma$  can be written as a product of  $m$  disjoint cycles of lengths  $k_1, \dots, k_m$ . For example, for  $\sigma = (123)(45)(6) \in S_6$ , we have  $m = 3$ ,  $k_1 = 3, k_2 = 2, k_3 = 1$ . Using this notation, we have  $\text{sign}(\sigma) = (-1)^{k_1-1+k_2-1+\dots+k_m-1} = (-1)^{n-m}$ . Thus  $\sigma$  is an even permutation (namely,  $\sigma \in A_n$ ) if and only if  $n-m$  is even. For example,  $(123)(45)(6) \in S_6$  has signature  $-1$ . This is Ex 10.1, page 74. Now suppose that  $n-m$  is even and so that  $\sigma \in A_n$ . We consider the  $S_n$  conjugacy class  $C(\sigma) = \{g\sigma g^{-1} : g \in S_n\}$ .

**Problem 3.** (1) If all  $k_i$  are odd and distinct, then  $C(\sigma)$  is the union of two  $A_n$  conjugacy classes and these two conjugacy classes have the same order.  
(2) Otherwise (which means, either one of  $k_i$  is even, or there are at least two  $k_i$  are the same), then  $C(\sigma)$  is still a single  $A_n$ -conjugacy class.

For example, in  $S_4$ , if  $\sigma = (12)(34)$ , then  $C(\sigma)$  is a single  $A_4$  conjugacy class; if  $\sigma = (123)(4)$ , then  $C(\sigma)$  is the union of two different  $A_4$  conjugacy classes. Actually, one can see that  $(123)$  and  $(132)$  are not conjugate in  $A_4$ .

**Problem 4.** Given an element  $p \in S_n$  with  $m_1$  1-cycles,  $\dots$ ,  $m_n$   $n$ -cycles. So  $\sum_{i=1}^n im_i = n$ . For example, for the cycle  $p = (123)(45)(67)(89)$  of  $S_{10}$ , we have  $m_1 = 1, m_2 = 3, m_3 = 1$  and  $m_i = 0$  for  $i \geq 4$ . Determine how many elements are in the conjugacy class determined by  $p$ .

Answer:

$$|C| = \frac{n!}{1^{m_1}m_1!2^{m_2}m_2!\dots n^{m_n}m_n!} = \frac{n!}{\prod_{i=1}^n i^{m_i}m_i!}.$$

For example, in  $S_5$ , the conjugacy class containing  $(12)(345)$  has  $\frac{5!}{2^1 \cdot 1! \cdot 3^1 \cdot 1!} = 20$  elements, and the conjugacy class containing  $(12345)$  has  $\frac{5!}{5^1} = 4! = 24$  elements.

**Problem 5.** Let  $G$  be a finite group,  $H$  be a subgroup of  $G$ . Let  $C \subset G$  be a conjugacy class and suppose

$$H \cap C = \coprod_{i=1}^r D_i,$$

where each  $D_i$  is a conjugacy class of  $H$ . Consider the set

$$X_i = \{(c, g) \in C \times G : g^{-1}cg \in D_i\}.$$

Express  $|X_i|$  in terms of  $|G|, |H|, |D_i|$ .

Hint: Consider the group action  $G \times X_i \rightarrow X_i$  defined by  $x.(c, g) = (xcx^{-1}, xg)$ . For  $d \in D_i$ , we consider  $(d, 1) \in X_i$  and its  $G$ -orbit  $O((d, 1))$ . If  $(c, g) \in X_i$ , then  $(c, g) = g.(g^{-1}cg, 1)$  and  $g^{-1}cg \in D_i$ . Thus  $X_i = \cup_{d \in D_i} O((d, 1))$ . Moreover, if  $O((d, 1)) = O((d', 1))$ , then  $d = d'$ . Thus  $X_i = \coprod_{d \in D_i} O((d, 1))$ . Consider the  $G$  action on  $O((d, 1))$  we have  $\text{Stab}((d, 1)) = 1$ . Thus  $|O((d, 1))| = |G|$  and  $|X_i| = |G||D_i|$ .

**Problem 6.** Let  $G = D_4 = \{1, x, x^2, x^3, y, xy, x^2y, x^3y\}$  with  $x^4 = 1 = y^2, yxy^{-1} = x^3$  and  $H = \{1, x^2, y, x^2y\} \subset G$ . Find all conjugacy classes  $C$  of  $G$ , and for each conjugacy class  $C$  of  $G$ , decompose  $C \cap H$  into conjugacy classes of  $H$ .

**Problem 7.** Let  $G = \text{GL}_2(\mathbb{F}_p)$ ,  $H = \text{SL}_2(\mathbb{F}_p) = \{g \in G : \det(g) = 1\}$ . Let  $C \subset G$  be the conjugacy class of the element  $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Namely,

$$C = \{gug^{-1} : g \in G\}.$$

Try to decompose  $C \cap H$  into conjugacy classes of  $H$ .

**Problem 8.** Let  $p$  be a prime number. Show that the cyclic group  $C_{p^n}$  for  $n \geq 2$  is not a semi-direct product of two proper subgroups.

Proposition 7.3.3, page 198, says that every group of order  $p^2$  is abelian. In the following, we give some examples of non-abelian group of order  $p^3$ . We assume that  $p > 2$ . If  $p = 2$ , we have seen that the quaternion group is an order 2 non-abelian group.

The first one is called Heisenberg group of the field  $\mathbb{F}_p$ , and we temporarily denote it by  $H(\mathbb{F}_p^2)$ . (This looks like a weird notation, but it has generalizations). It is defined by

$$H(\mathbb{F}_p^2) = \left\{ \begin{bmatrix} 1 & x & z \\ & 1 & y \\ & & 1 \end{bmatrix} \in \text{GL}_3(\mathbb{F}_p), x, y, z \in \mathbb{F}_p \right\}.$$

Its group structure is defined by matrix multiplication. The other group is temporarily denoted by  $G_p$  and it is defined by

$$G_p = \left\{ \begin{bmatrix} x & y \\ & 1 \end{bmatrix} \in \text{GL}_2(\mathbb{Z}/p^2\mathbb{Z}) : x \equiv 1 \pmod{p}, y \in \mathbb{Z}/p^2\mathbb{Z} \right\}.$$

**Problem 9.** Show that  $H(\mathbb{F}_p^2)$  and  $G_p$  are non-abelian group of order  $p^3$ . Moreover, show that they are indeed semidirect products of their own subgroups.

It can be shown that these are the only two non-abelian groups of order  $p^3$  up to isomorphism.

**Problem 10.** Determine the class equations of  $D_n, n \geq 3$ ,  $\text{GL}_2(\mathbb{F}_p), \text{SL}_2(\mathbb{F}_p)$  and  $\text{PSL}_2(\mathbb{F}_p)$ , where  $\text{PSL}_2(\mathbb{F}_p) = \text{SL}_2(\mathbb{F}_p) / \{\pm I_2\}$ .

If you find this hard, at least try some examples for small  $p$ .

**Problem 11.** Let  $G$  be a finite group and let  $p$  be the smallest prime that divides  $|G|$ .

- (1) If  $H$  is a normal subgroup of  $G$  such that  $|H| = p$ , show that  $H < Z(G)$ .
- (2) If  $H < G$  is a subgroup such that  $[G : H] = p$ , then  $H$  is a normal subgroup of  $G$ .

Part (1) is Ex 6.4 page 223. and part (2) generalizes Ex.8.10, page 73.

## 1. OUTER AUTOMORPHISM OF $S_6$

**Problem 12.** Show that  $\text{PSL}_2(\mathbb{F}_5) \cong A_5$  and  $\text{PGL}_2(\mathbb{F}_5) \cong S_5$ .

As in the proof that the icosahedral group is isomorphic to  $A_5$ , one needs to find a set with 5 elements such that  $\text{PSL}_2(\mathbb{F}_5)$  acts on it. This is Exercise 8.4, page 287.

Let  $X = \mathbb{P}^1(\mathbb{F}_5)$ . We know that  $\text{PSL}_2(\mathbb{F}_5)$  and  $\text{PGL}_2(\mathbb{F}_5)$  act on  $X$  in a natural way. See last HW. Thus we get a group homomorphism  $f : \text{PSL}_2(\mathbb{F}_5) \rightarrow S_6 \cong \text{Perm}(X)$  and  $g : \text{PGL}_2(\mathbb{F}_5) \rightarrow S_6 \cong \text{Perm}(X)$ . From last HW, we know  $f$  is injective and its image is a transitive subgroup of  $S_6$ . Denote  $H := \text{Im}(f)$ , which is a subgroup of  $S_6$ .

**Problem 13.** Let  $Y$  be the set of all conjugates of  $H$ . Namely,  $Y = \{gHg^{-1} \mid g \in S_6\}$ .

- (1) Show that  $N_{S_6}(H) = \text{Im}(g) \cong S_5$  and conclude that  $|Y| = 6$ .
- (2) Consider the action of  $S_6$  on  $Y$  by conjugation. Namely  $S_6 \times Y \rightarrow Y$ ,  $(g, xHx^{-1}) \mapsto gxHx^{-1}g^{-1}$ . This gives a homomorphism  $\Phi : S_6 \rightarrow \text{Perm}(Y) \cong S_6$ . Show that  $\Phi$  is an isomorphism.
- (3) Show that  $\Phi$  is not an inner automorphism.

It is a fact that  $\text{Aut}(S_6)$  is generated by inner automorphisms and  $\Phi$ .